

Design Pattern 5: Decorator Pattern

Exercise 5: Implementing the Decorator Pattern

Scenario:

You are developing a notification system where notifications can be sent via multiple channels (e.g., Email, SMS). Use the Decorator Pattern to add functionalities dynamically.

Steps:

Create a New Java Project:

Create a new Java project named DecoratorPatternExample.

Define Component Interface:

Create an interface Notifier with a method send().

Implement Concrete Component:

Create a class EmailNotifier that implements Notifier.

Implement Decorator Classes:

Create abstract decorator class NotifierDecorator that implements Notifier and holds a reference to a Notifier object.

Create concrete decorator classes like SMSNotifierDecorator, SlackNotifierDecorator that extend NotifierDecorator.

Test the Decorator Implementation:

Create a test class to demonstrate sending notifications via multiple channels using decorators.